Two independent finite-difference solvers for an open Helmholtz resonator: verification and comparison with published measurements

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October 2026

Abstract

The natural frequency of a Helmholtz resonator depends on an end correction that lumped models assume. I compute it instead with two independent axisymmetric finite-difference solvers for an open-neck resonator: a frequency-domain solver that closes the mouth with the radiation impedance of a baffled piston, and a time-domain leapfrog solver that meshes the exterior half-space and lets the resonator select its own frequency after a broadband pulse. Manufactured solutions give observed orders of 2.03 and 1.98; the time-domain test also exposed a half-cell offset of the rigid walls, an $O(h)$ geometric error that is removed here by Richardson extrapolation. The two solvers then give 203.4 and 204.6 Hz, with grid uncertainties (GCI) of 2.0% and 1.5% and a further 0.46% from the outer boundary of the time-domain solver; they differ by 0.6%. A study of three outer boundaries shows that the original sponge layer reflected 44% of the incident wave, which explained a factor-six spread in the computed radiation Q. Compared with published measurements, the solver is within 0.9% and 2.2% of two resonators of Selamet et al. with no fitted parameter, and is 10–13% above an Erlenmeyer flask. The solvers are lossless, so damping is not predicted. A training-free test shows that neural bases fit the field to better than 0.4% while the PDE residual of that fit is 2,536–9,979 times that of the zero field.

1 Introduction

A Helmholtz resonator is a cavity of volume V connected to the outside through a neck of length L and cross-section $S = \pi a^2$. The air plug in the neck oscillates against the compressibility of the air in the cavity, and the lossless lumped model gives

$$f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{V\,(L + \Delta L)}}, \tag{1}$$

where ΔL, the end correction, accounts for the air that moves with the plug just outside each end of the neck [7, 8, 21]. ΔL is of the order of the neck radius and is usually taken from tables. For a neck as short as the one studied here ($L = 4a$), it lowers f_0 by about 15%.

A numerical model can compute ΔL instead of assuming it, but only if its own error is known. The question addressed here is therefore narrow: can a finite-difference model of an open resonator predict f_0 with a stated numerical uncertainty, and where does it stop agreeing with measurements? The contributions are:

*This work started as the author's individual TIPE (the research project of the French *classes préparatoires*) in 2023 and was rebuilt and extended in 2026. Code, data and the scripts that produce every figure and number: `https://github.com/clementdumeril/helmholtz-resonator-solver`.

- two independent axisymmetric finite-difference solvers, one in the frequency domain with an impedance condition at the mouth and one in the time domain with a meshed exterior and three interchangeable outer boundaries (sponge, first-order Mur, PML);
- a verification sequence (manufactured solutions, Richardson extrapolation with a grid convergence index, cross-solver comparison, outer-boundary study) in which the manufactured solution revealed a half-cell offset of the rigid walls;
- a comparison with three published measured resonators, two matched and one missed, and a training-free measurement of how well neural bases represent the computed field and its PDE residual (Appendix A).

2 Related work

End corrections and measured resonators. Rayleigh derived the end correction of a flanged pipe [8]; Ingard studied the theory and design of resonators, including the end corrections on both sides of the neck [7]; Morse and Ingard give the radiation impedance of a baffled piston used here [10]. Selamet et al. studied concentric cylindrical resonators with two-dimensional analytical models, a three-dimensional boundary-element method and experiments [19]; two of their configurations are used in Section 6.2. Moloney measured quality factors of glass bottles and found them slightly lower than his predictions [9]. Indenbom and Pogossian measured the resonance of a filled Erlenmeyer flask [6]; its empty case is used here.

Verification. The distinction between code verification, solution verification and validation follows Roache [17] and Oberkampf and Roy [13]. Code verification uses the method of manufactured solutions [18]. Discretisation uncertainty is reported with the grid convergence index (GCI) [2, 16].

Open boundaries. The first-order absorbing condition of Mur [12] is exact only at normal incidence. The perfectly matched layer (PML) of Bérenger [1] is reflectionless in the continuous problem for a planar interface; on curved boundaries it requires a curvilinear stretch [3]. The time-domain PML used here is the unsplit second-order form for the scalar wave equation with two auxiliary fields [4].

Neural PDE solvers. Physics-informed neural networks minimise the PDE residual over the parameters of a network [15]. Their difficulties on oscillatory problems are often attributed to spectral bias [14, 22], with remedies such as sine activations [20] and domain decomposition [11]. Trefftz methods use functions that satisfy the homogeneous Helmholtz equation exactly [5].

3 Problem formulation

Geometry. The resonator is axisymmetric (Table 1): a cylindrical neck of radius $a = 10$ mm and length $L = 40$ mm opens at $z = 0$ into a rigid baffle and joins a cylindrical cavity of radius $R_c = 40$ mm and height 80 mm. All walls are rigid. The air is lossless, with $c = 343$ m/s and $\rho = 1.204$ kg/m^3.

Equations. In the time domain the acoustic pressure obeys

$$\partial_t^2 p = c^2 \nabla^2 p + F(\mathbf{x}, t), \qquad \nabla^2 p = \partial_r^2 p + \frac{1}{r}\partial_r p + \partial_z^2 p, \tag{2}$$

Table 1: Reference geometry and numerical parameters.

	quantity	value
geometry	neck radius a / length L	10 / 40 mm
	cavity radius R_c / height	40 / 80 mm
air	c, ρ	343 m/s, 1.204 kg/m^3
frequency domain	volume source (Gaussian, cavity centre)	width 10 mm
	peak search	0.05 Hz steps, 3-point parabola
time domain	exterior (radius × depth)	120 × 100 mm
	absorbing layer	sponge, 30 mm
	source	Ricker, 250 Hz centre, at $z = -55$ mm
	time step	$\Delta t = 0.5\,h/(c\sqrt{2})$
	simulated time; fit window	250 ms; $t \geq 45$ ms
grids	h	2, 1, 0.5 (0.25) mm

and in the frequency domain, at angular frequency ω and $k = \omega/c$,

$$\nabla^2 p + k^2 p = -F(\mathbf{x}). \tag{3}$$

On the axis, $(1/r)\,\partial_r p$ is indeterminate; since $\partial_r p = 0$ there, L'Hôpital's rule gives $(1/r)\,\partial_r p \to \partial_r^2 p$, so $\nabla^2 p = 2\,\partial_r^2 p + \partial_z^2 p$ at $r = 0$. Rigid walls impose $\partial_n p = 0$.

Radiation. The two solvers model the exterior differently, so their agreement tests the radiation model. The frequency-domain solver stops at the mouth and applies the low-frequency impedance of a baffled piston [10],

$$Z_r = \rho c \left[\tfrac{1}{2}(ka)^2 + \mathrm{i}\,\tfrac{8}{3\pi} ka\right], \tag{4}$$

as a Robin condition $\partial_z p = \alpha p$ on $z = 0$, $r \leq a$, with $\alpha = \mathrm{i}\omega\rho/Z_r$. Its reactance contains the exterior end correction $8a/(3\pi)$, so this solver imposes the exterior part of ΔL. The time-domain solver meshes a baffled exterior half-space and computes the whole end correction; its outer boundary has to absorb the outgoing wave.

Reference values. Two lumped references are used, and neither is a measurement. (i) $\Delta L = 0.85a + 0.66a = 1.51a$, the flanged exterior correction [8] plus an interior correction of $0.66a$, which gives $f_0 = 205.6$ Hz. The $0.66a$ is close to the $0.67a$ fitted with this solver's own Dirichlet-mouth runs (Appendix B), so agreement with this reference is partly circular. (ii) Ingard's interior correction $0.85a\,(1 - 1.25\,a/R_c) = 0.58a$ [7] with the same exterior term gives $\Delta L = 1.43a$ and $f_0 = 207.0$ Hz, independent of this work.

4 Numerical method

Both solvers use a uniform grid of step h on the meridian plane (Fig. 1) and the standard second-order stencil of the axisymmetric Laplacian, with the radial part written as $4(p_1 - p_0)/h^2$ on the axis.

Frequency domain. The operator of Eq. (3) is assembled as a sparse complex matrix. A rigid wall is imposed by folding the ghost node onto the diagonal: the coefficient of the missing neighbour is added to the diagonal instead of appearing off-diagonal, which enforces a zero flux across the face half a cell beyond the last fluid node. The Robin condition at the mouth eliminates the ghost node through $p_{-1} = p_1 - 2h\alpha p_0$. Because k^2 and α depend on ω, each frequency requires a new sparse LU factorisation (SciPy): 0.1 s on the 1 mm grid (4 961 nodes), 4 s on the 0.25 mm grid (77 441 nodes). A Gaussian volume source of width 10 mm drives the

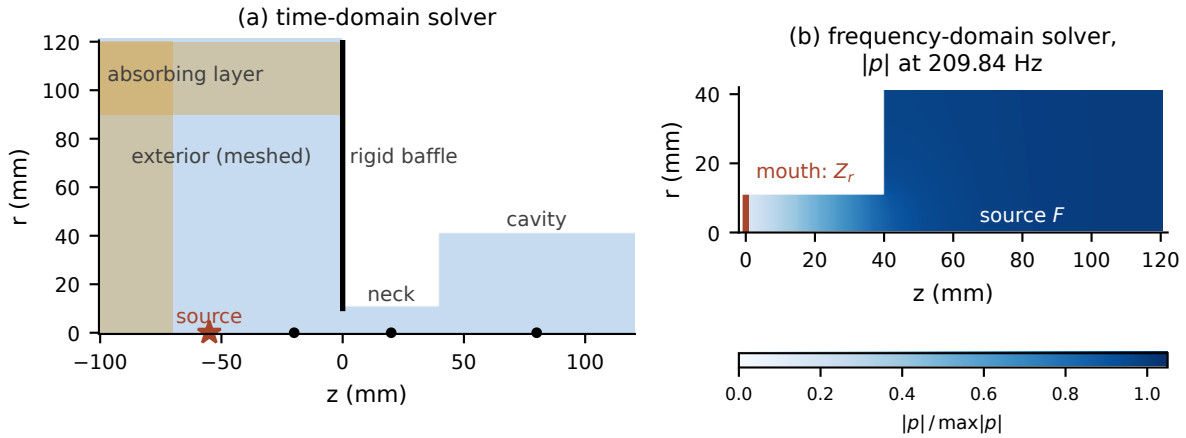


Figure 1: Computational domains on the meridian plane. (a) Time-domain solver: open neck, meshed exterior with a rigid baffle at $z = 0$, absorbing layer (shaded) on the outer boundaries, Ricker source (star) and the three probes (dots). (b) Frequency-domain solver: the domain stops at the mouth, which carries the radiation impedance; colour shows $|p|$ at 209.84 Hz on the 0.5 mm grid.

cavity; f_0 is the peak of the cavity pressure, located by a three-point parabola on a 0.05 Hz sweep, and the radiation Q is f_0 over the -3 dB width. A second configuration of the same scheme, with the mouth set to $p = 1$ (Dirichlet), is used for the neck-length study and for Selamet's side-branch resonators.

Time domain. Equation (2) is advanced by explicit leapfrog with $\Delta t = 0.5\,h/(c\sqrt{2})$. The Laplacian is written in masked finite volumes: fluxes through faces between a fluid and a solid cell are zero, which again places each wall on the face half a cell beyond the last node. The neck opens onto an exterior of radius 120 mm and depth 100 mm terminated by 30 mm sponge layers. A Ricker pulse centred at 250 Hz is emitted 55 mm in front of the mouth, so the resonator is excited over a broad band and rings at its own frequency after the pulse has passed. f_0 and Q are obtained by fitting $Ae^{-t/\theta}\cos(2\pi f_0 t + \phi)$ to the cavity probe for $t \geq 45$ ms, with $Q = \pi f_0 \theta$. The fit locates f_0 well below the 4.9 Hz resolution of an FFT over the same window.

Outer boundaries. Three terminations share one solver core: (i) a sponge, i.e. a damping term $\sigma\,\partial_t p$ with $\sigma = \sigma_{\max} d^2$ across the layer, as in the solver above; (ii) the first-order Mur condition $\partial_t p + c\,\partial_n p = 0$ on the outer ring [12]; (iii) the unsplit PML [4]

$$\begin{aligned} \partial_t^2 p + (s_r + s_z)\,\partial_t p + s_r s_z\, p &= c^2\nabla^2 p + \nabla\cdot\boldsymbol{\psi}, \\ \partial_t \psi_r + s_r \psi_r = c^2(s_z - s_r)\,\partial_r p, &\qquad \partial_t \psi_z + s_z \psi_z = c^2(s_r - s_z)\,\partial_z p, \end{aligned} \tag{5}$$

with quadratic profiles s_r, s_z. This is the Cartesian stretch applied in the (r, z) plane: the $1/r$ term of a cylindrical PML [3] is left out.

5 Verification

5.1 Code verification: manufactured solutions

For the frequency-domain scheme, the solution $p_e = \cos(\pi r/R)\cos(\pi z/H)$ on a 40×120 mm rectangle, regular on the axis, is imposed through its source term at the wavenumber of 235 Hz. Over four grids from 4 to 0.5 mm the RMS error decreases at order 2.03 (Fig. 2a). This test uses Dirichlet edges and therefore does not exercise the wall or mouth treatment.

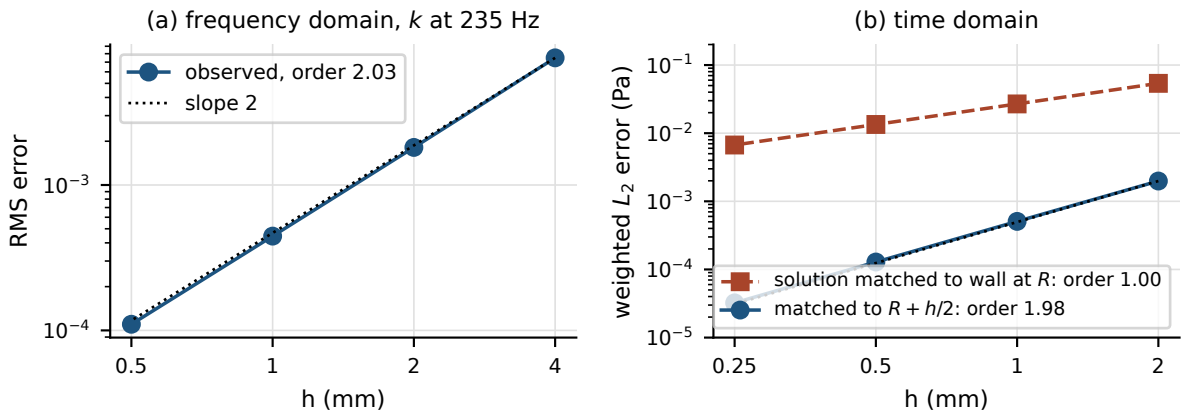


Figure 2: Manufactured solutions. (a) Frequency-domain scheme on a smooth domain with Dirichlet edges. (b) Time-domain scheme with rigid walls: matched to the nominal wall position the error converges at first order; matched to the wall at $R + h/2$ it converges at second order.

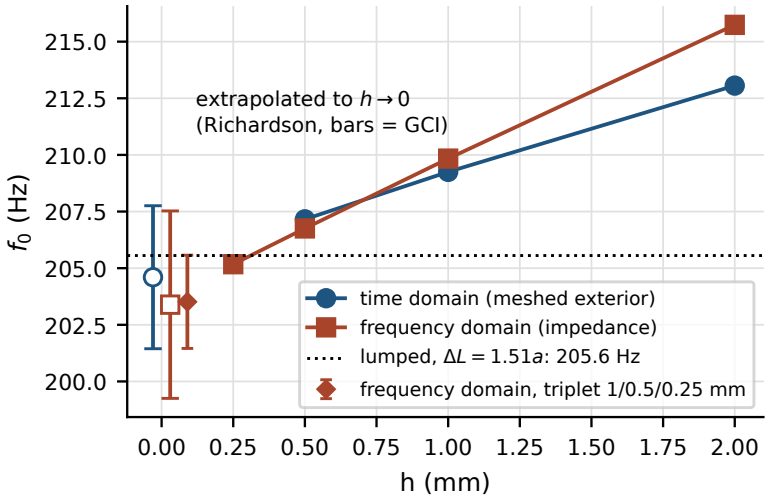


Figure 3: Natural frequency of the reference resonator against grid step for both solvers, with the Richardson extrapolations at $h = 0$ and their GCI, and the semi-empirical lumped reference (i).

The time-domain test does exercise the walls. It uses $p = J_0(\lambda r)\cos\big(k_z(z + z_0)\big)\,T(t)$ in a closed cylinder, with λ chosen so that $\partial_r p = 0$ on the lateral wall and k_z so that $\partial_z p = 0$ on both ends, and $T(t) = u^3 e^{-u}$, $u = t/\tau$, with τ matched to the wave period so that $\partial_t^2 p$ and $c^2\nabla^2 p$ are of the same size. When the manufactured solution places the walls at the nominal radius R and height H, the observed order is 1.00 on four grids (Fig. 2b). When it places them at $R + h/2$ and $H + h$, the observed order is 1.98 and the error at $h = 2$ mm is 27 times smaller. The interior scheme is second order; the first-order error comes from the wall position. The frequency-domain scheme has the same convention through its ghost-node folding. Both solvers therefore simulate a resonator whose dimensions are enlarged by $O(h)$, and the full discretisation of the resonator is first order in h by construction. Aligning the cell faces with the nominal walls would remove this term and should restore second order; it was not done here, and the offset is handled by extrapolation instead.

Table 2: Grid convergence of f_0 (Hz) on the reference geometry. Time domain: damped-sinusoid fit (sponge layer); frequency domain: peak of the response.

solver	$h = 2$	$h = 1$	$h = 0.5$	$h = 0.25$ mm	p	$f_{h\to 0}$	GCI
time domain (meshed exterior)	213.06	209.25	207.16	–	0.86	204.60	1.5%
frequency domain (impedance)	215.74	209.84	206.75	205.18	0.94	203.39	2.0%
triplet 1 / 0.5 / 0.25 mm					0.96	203.52	1.0%

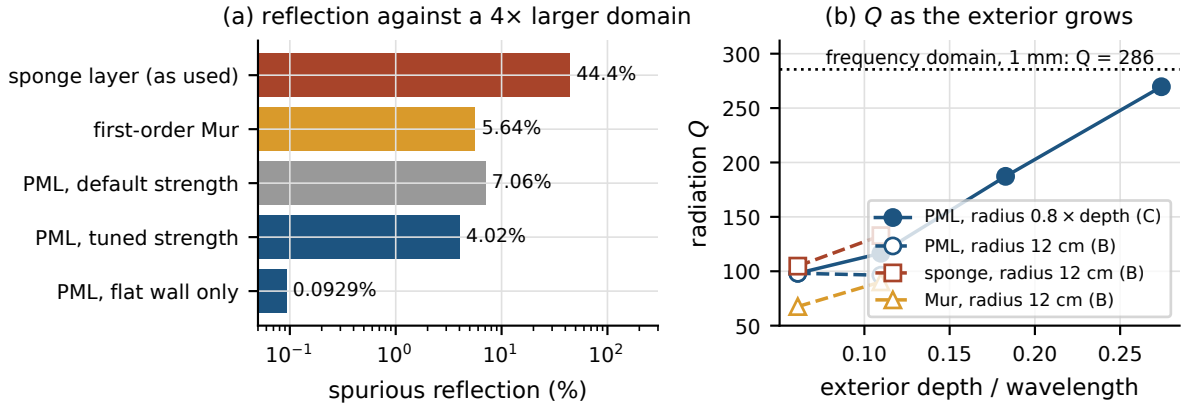


Figure 4: Outer boundaries. (a) Spurious reflection of a 4 kHz-centred Ricker pulse, measured at an off-axis probe against the same run on a domain four times larger. (b) Radiation Q of the resonator (1 mm grid) against exterior depth. Configuration B (open symbols) changes only the depth, at a fixed radius of 12 cm; configuration C (filled) grows the radius with the depth (0.8× depth, at least 12 cm). Dotted line: Q of the frequency-domain solver with the analytic impedance, 1 mm grid.

5.2 Solution verification: grid convergence

On three grids with refinement ratio 2, the observed order p, the extrapolated value and the fine-grid GCI are [2, 16]

$$p = \frac{\ln\left|(f_3 - f_2)/(f_2 - f_1)\right|}{\ln 2}, \quad f_{h\to 0} = f_1 + \frac{f_1 - f_2}{2^p - 1}, \quad \mathrm{GCI} = 1.25\,\frac{|f_2 - f_1|/f_1}{2^p - 1}, \tag{6}$$

with f_1 on the finest grid. Since the expected order is 1, successive differences should halve with each refinement. The ratios of successive differences are 1.92 and 1.95 for the frequency-domain solver and 1.82 for the time-domain solver, close to 2, so the three-grid GCI with a safety factor of 1.25 is applicable.

Table 2 and Fig. 3 give the results. For the frequency-domain solver a fourth grid at 0.25 mm gives a second triplet with $p = 0.96$; it moves the extrapolated value by 0.06%, well inside the GCI. A 0.25 mm time-domain run would take about 8 h on one core and was not done, so the time-domain extrapolation rests on three grids.

Uncertainty budget. The time-domain runs of Table 2 use the sponge layer. With the same grid (1 mm) and the three terminations of Section 5.3, f_0 spreads over 0.46%, which is added to its uncertainty. The two extrapolated values, $204.60 \pm 1.5\%$ (grid) $\pm 0.46\%$ (outer boundary) and $203.39 \pm 2.0\%$ (grid), differ by 0.59%, inside both bands. Their raw 1 mm values agree to 0.3%, but both carry the same wall offset, so that agreement alone says little; only the extrapolated values are compared.

Table 3: Outer-boundary treatments. Reflection: peak spurious signal relative to the peak incident signal. Drift: change in the radiation Q of the resonator (1 mm grid) when the exterior depth goes from 10 to 18 cm at a fixed radius of 12 cm (configuration B).

treatment	spurious reflection	Q drift, depth $10 \to 18$ cm, radius 12 cm
sponge layer (original solver)	44%	+26.8%
first-order Mur	5.6%	+33.5%
PML (tuned strength)	4.0%	−1.8%

5.3 Outer boundary of the time-domain solver

With the original sponge layer, the radiation Q ranged from 44 to 267 as the layer strength, thickness and domain size were changed on the 2 mm grid. (The f_0 recorded in that study came from an FFT with a 4 Hz bin and cannot show whether the frequency moved.) To find out why, each termination was tested on an empty exterior box: a pulse is fired towards the boundary and the probe signal is compared with the same run on a domain four times larger, from which nothing can return within the 1.4 ms window. The difference is the spurious reflection, with no analytic reference needed.

The sponge reflected 44% of the incident amplitude (Table 3, Fig. 4a), enough to explain the spread in Q. The PML reflects 7.1% at the default strength and 4.0% after scanning it. Running each wall alone isolates its limit: 0.093% (−61 dB) on the flat boundary $z = \text{const}$, where the Cartesian stretch is exact, and 4.0% (−28 dB) on the cylindrical boundary, where the $1/r$ term is missing. Neither a stronger profile nor a thicker layer lowers that floor.

The resonator was then run in two configurations (Fig. 4b). In configuration B only the depth changes, from 10 to 18 cm, at a fixed radius of 12 cm. The PML Q moves by −1.8%, against +26.8% for the sponge and +33.5% for the Mur condition. This only shows that Q is insensitive to the position of the flat boundary, where the PML is nearly exact. In configuration C the radius grows with the depth (12, 14.4, 24 and 36 cm). At 18 cm depth, moving the cylindrical boundary from 12 to 14.4 cm raises the PML Q from 96.4 to 116.6 (+21%): the curved boundary, which carries the 4% floor, still controls the damping. Over configuration C, with the exterior growing from 0.061 to 0.110, 0.183 and 0.274 wavelengths (the wavelength is 1.64 m at 209 Hz), Q goes from 98.1 to 116.6, 187.2 and 269.6, towards the $Q = 285.6$ of the frequency-domain solver on the 1 mm grid (289.7 and 290.5 on the 0.5 and 0.25 mm grids). The time-domain radiation Q is therefore not converged. The frequency moves less: 0.46% across the three terminations in configuration B and 0.10% over configuration C.

6 Results

6.1 The resonance and the end correction

In the time domain the resonator selects its frequency: after the pulse has passed, the cavity rings at f_0 with no frequency imposed anywhere (Fig. 5a). In the frequency domain the response is narrow, with a −3 dB width of 0.73 Hz on the 1 mm grid (Fig. 5b).

Inverting Eq. (1) gives the end correction implied by each frequency. From the raw 1 mm time-domain value it is $1.32a$, 13% below the $1.51a$ of reference (i); from the extrapolated value it is $1.56a$. The 13% deficit at 1 mm is the wall offset. The extrapolated value is 3% above reference (i), which is partly circular, and 9% above Ingard's $1.43a$ (reference (ii)); in frequency, both extrapolated solvers lie 1.2–1.7% below the Ingard value of 207.0 Hz. With the impedance condition, the frequency-domain study gives a shift in ΔL of $0.823a$ between the Dirichlet and the radiating mouth, against $8a/(3\pi) = 0.849a$ in Eq. (4). Since that reactance is imposed, this is a consistency check of the chain, not an independent computation.

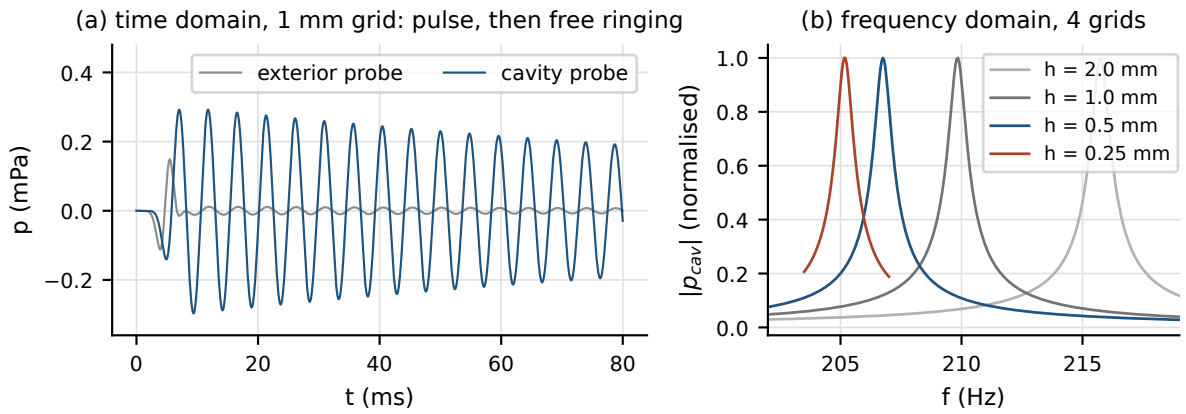


Figure 5: (a) Time-domain solver, 1 mm grid: the pulse reaches the exterior probe, then the cavity keeps oscillating at its own frequency. (b) Frequency-domain response of the cavity on successive grids; the peak moves down as the walls move in.

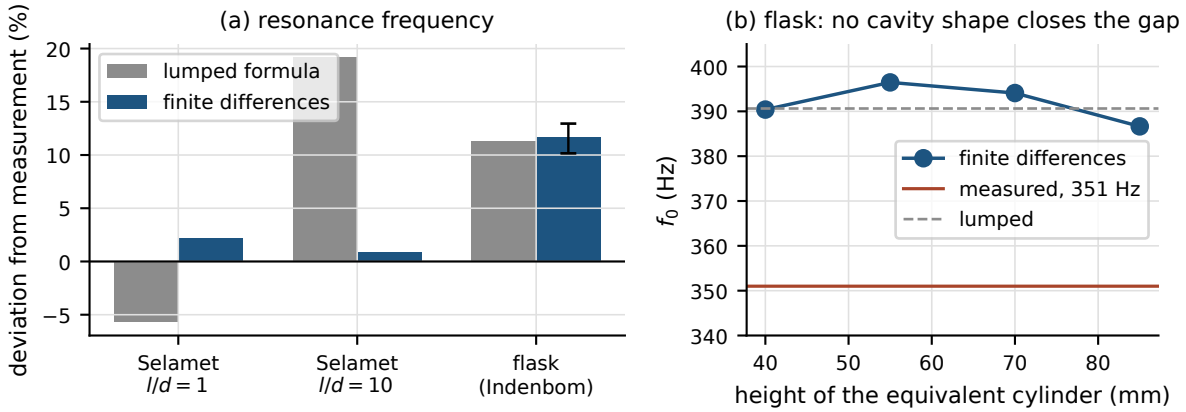


Figure 6: Comparison with published measurements. (a) Deviation of the lumped formula and of the finite-difference solver; the bar on the flask spans the cavity aspect ratios tried. (b) Flask: finite-difference f_0 against the height of the equivalent cylindrical cavity.

6.2 Comparison with published measurements

Selamet et al. measured two concentric resonators with the same 4500 cm^3 cavity in two shapes ($l/d = 1$ and 10) and a neck of diameter 40.44 mm and length 85 mm [19]. They are side branches on a duct, and were computed with the Dirichlet mouth. The lumped formula knows only the volume and returns 85.8 Hz for both, −5.7% and +19% from the measurements. The two-dimensional field is within 2.2% and 0.9% of both (Table 4). This agreement should not be over-read. The values come from the raw 1 mm grid, which on the reference geometry sits about 3% above the extrapolated frequency, and the Dirichlet mouth leaves out the end correction on the duct side of the neck. Both effects raise the computed frequency, so a converged computation with the duct included would land lower, possibly below the measurements; the present agreement may be partly fortuitous.

Indenbom and Pogossian measured an empty 100 mL Erlenmeyer flask at 351 Hz [6], where the lumped formula gives 391 Hz; my lumped model gives 390.6 Hz. The flask is conical and my cavity is a cylinder, so I replaced it by a cylinder of equal volume and swept its aspect ratio. The solver stays between 387 and 396 Hz for every shape (Fig. 6b), 10–13% above the measurement, so resolving the field near the neck does not close the gap. In terms of Eq. (1), the measurement corresponds to an effective neck of about 54 mm instead of 43 mm. The cause is not established;

Table 4: Resonance frequencies (Hz) against published measurements. Finite differences on the 1 mm grid, not extrapolated, no fitted parameter.

case	measured	lumped, Eq. (1)	finite differences	deviation
Selamet et al., $l/d = 1$	91	85.8	93.0	+2.2%
Selamet et al., $l/d = 10$	72	85.8	72.7	+0.9%
Indenbom & Pogossian, flask	351	390.6	387–396	+10 to +13%

the conical shoulder, which this solver cannot represent, is one candidate.

For damping, the same paper gives a half-width that brackets the measured Q between 18 and 30 (the paper does not state whether the width is taken at half amplitude or half power). An estimate combining viscothermal losses in the neck and radiation gives about 40: the model is optimistic, in the same direction as Moloney's comparison [9]. Neither solver contains viscothermal losses.

6.3 Representing the field with neural bases

Appendix A projects the frequency-domain field onto 96 functions from each of five neural bases and a Trefftz basis, without training. The projections fit the field to 0.15–0.36%, but their PDE residual is 2,536 to 9,979 times that of the zero field. The fits that minimise the residual instead reach 0.8–19% of the zero-field residual but miss the field entirely (field error of 100% or more). These numbers concern projections of a discrete field onto fixed random bases; no network is trained.

7 Discussion and limitations

What the results support. Two solvers with different discretisations and different radiation models agree on the natural frequency of the reference resonator to 0.6%, inside their uncertainties of about 2%. The verification sequence found an error that the cross-check alone would have hidden: both solvers carry the same half-cell wall offset, so their raw values agree better than either agrees with its own converged value.

What they do not support.

- *Damping.* Both solvers are lossless. The viscothermal boundary layer in the neck (about 0.14 mm at this frequency) is the dominant loss and is only estimated analytically. The Q values above are radiation-only and are not predictions of a real resonator.
- *Grid convergence.* The full discretisation is first order because of the wall offset and the staircase walls. The time-domain extrapolation rests on three grids and carries an extra 0.46% from the outer boundary.
- *Reference values.* The semi-empirical $1.51a$ uses an interior correction close to this solver's own fit; against Ingard's independent value the end correction differs by 9%.
- *Geometry.* Cavities are cylinders. The flask miss is not explained.
- *Outer boundary.* The Cartesian PML has a 4% reflection floor on the curved boundary, the reflection was measured with a 4 kHz-centred pulse whose wavelength is much shorter than at the resonance, and the radiation Q is not converged in exterior size.
- *Comparison data.* All measurements are other authors'. The two matched cases are side-branch resonators computed with a Dirichlet mouth on a single grid, so their agreement may be partly fortuitous.

8 Conclusion and future work

A finite-difference model of an open Helmholtz resonator predicts its natural frequency with a stated uncertainty of about 2%, provided that the half-cell wall offset revealed by the manufactured solution is accounted for, here by extrapolation. The end correction computed by the meshed exterior, $1.56a$, lies between the semi-empirical value (+3%) and Ingard's (+9%). Two published resonators are matched to 0.9–2.2% under the caveats above, one is missed by about 11%, and damping is outside the scope of these lossless models. The next steps are a wall treatment aligned with cell faces to recover second-order convergence, a cylindrical PML to remove the 4% floor, a converged side-branch computation for Selamet's resonators, a viscothermal boundary-layer model in the neck, and a ring-down measurement of a real resonator, for which the protocol and analysis script are in the repository.

Reproducibility statement

All code, data and figures are in the repository. The figures and the numbers quoted in the text are produced by `pythonpaper/make_figures.py` from versioned files in `data/`, and the report is built by `paper/build_paper.sh`. Hardware and software: Intel Core i7-6700HQ (4 cores), 32 GB RAM, Windows 10; Python 3.11.9, NumPy 2.2.6, SciPy 1.15.2, Matplotlib 3.10.6, PyTorch 2.8.0 (CPU); one core per run. Code map, with approximate run times:

- frequency-domain solver: `src/harmonic_solver.py`; sweeps with `studies/frequency_sweep.py` and `SW_DF=0.05`, over 195–220 Hz for `SW_H` = 2, 1, 0.5 (under 1, 1 and 6 min) and over 203.5–207 Hz for `SW_H=0.25` (71 solves, 9 min); four-grid analysis `verification/control_grid.py`;
- time-domain solver: `src/transient_solver.py` with `OR_TMS=250` and `OR_H` = 2, 1, 0.5 (about 1, 5 and 40 min); grid convergence `verification/grid_convergence.py`;
- manufactured solutions: `verification/mms_transient.py` (15 s) and section 3 of `notebooks/harmonic_study.ipynb`;
- outer boundaries: `studies/radiation_boundaries.py` (part A a few minutes, parts B and C several hours);
- published measurements: `experiments/literature_validation.py` (about 4 min) and section 5 of the notebook;
- neural bases: `studies/neural_basis_benchmark.py` (about 1 min);
- neck-length study: sections 4 and 6 of the notebook.

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Appendix A Representing the field with neural bases

Set-up. The target is the frequency-domain field on the 1 mm grid at 209.84 Hz, which is that grid's own resonance; on the 0.5 mm grid the same frequency is off resonance and the cavity response is about 8 times weaker. The target is therefore a discrete solution, not the converged field. Six families of 96 functions are taken at random initialisation: a tanh MLP, SIREN with $\omega_0 = 5$ and 8, Gabor atoms, a three-zone partition of unity of sine networks in the spirit of FBPINNs, and a Trefftz basis $J_0(\kappa_m r)Z_m(z)$. For each, two least-squares problems are solved: (i) the projection of the field onto the span, reporting its relative field error and its residual $\|\nabla^2\tilde{p} + k^2\tilde{p} + F\|/\|F\|$, computed by automatic differentiation on 900 points of the fluid domain; (ii) the fit that minimises that residual over the span, reporting the same two numbers. A residual of 1 is what the zero field achieves. No boundary terms are included, and no network is trained.

Results. The projections fit the field to 0.15–0.36% (Fig. 7a), with residuals 2,536 to 9,979 times that of the zero field (Fig. 7b). The minimum-residual fits reach 0.8–19% of the zero-field residual, but their field error is 100% or more: without boundary terms the interior residual does not determine the field, and neither fit comes close to satisfying both objectives. The conditioning of the residual operator varies by eleven orders of magnitude between families (from 5×10^2 to 7×10^{13}), with sine activations far better conditioned than tanh, and this does not change the picture. The Trefftz basis satisfies $\nabla^2\phi + k^2\phi = 0$ exactly, so both of its residuals equal that of the zero field; it cannot carry the source and fits the field only to 4.0%.

Interpretation. On the sample points $\|k^2 p\|/\|F\| \approx 49$, so a smooth error of 0.15% on p would contribute only about 7% of $\|F\|$ through the $k^2 p$ term. The residual is dominated instead by the Laplacian of the error: an error of relative size ε varying on a length ℓ gives a residual of order $\varepsilon\,(\|k^2 p\|/\|F\|)/(k\ell)^2$. For the best basis (Gabor) this matches the measured residual for $\ell \approx 1.4$ mm, of the order of the grid step. Much of the residual therefore plausibly comes from grid-scale structure of the discrete target (for example at the staircase walls), which no smooth basis reproduces. The test shows that projecting this discrete field onto these bases gives an unusable residual, and that minimising the interior residual alone does not recover the field; it does not show that the bases cannot represent the Laplacian of the continuous solution.

Appendix B Neck-length study with the Dirichlet mouth

With the Dirichlet mouth, the three-grid study over five neck lengths from 2 to 6 cm gives observed orders between 0.87 and 0.95 and GCIs between 2.0% and 2.3% (mean 2.1%; notebook section 4). A 0.25 mm grid for $L = 2$ and 4 cm moves the observed order from 0.91 to 1.00 and from 0.87 to 1.01, and the extrapolations by 0.20% and 0.36%. Fitted on the extrapolated frequencies, a free power law AL^b and the effective-length model $A/\sqrt{L + \Delta L}$ cannot be told apart (ΔAICc = 0.3); the effective-length fit gives $\Delta L = 0.67a$ and A close to the theoretical 48.25 $\mathrm{m}^{1/2}/\mathrm{s}$.

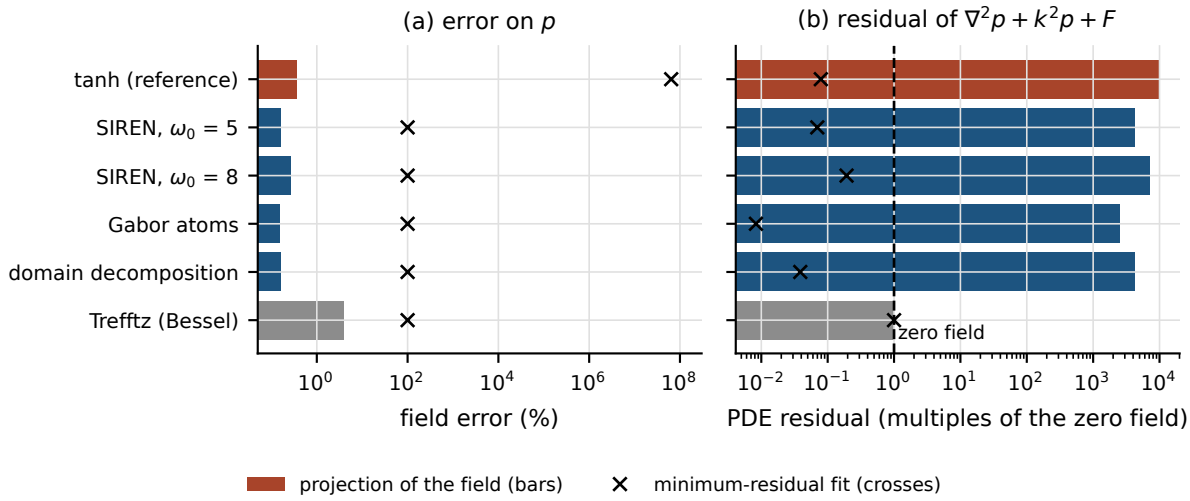


Figure 7: Least-squares fits with 96 functions of each family. Bars: projection of the field; crosses: the fit that minimises the PDE residual. (a) Relative error on p. (b) PDE residual in multiples of that of the zero field (dashed line).